

DE.AUTO

CHAPTER 8

SUSPENSION & HANDLING

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English



POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE MUST-KNOW TERMS

Car modifications, explained

Chapter Eight: Suspension and Handling

“GRIP IS A CONVERSATION BETWEEN CAR AND ROAD.”

— DE.AUTO

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You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

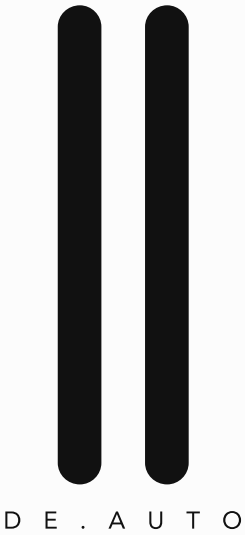
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

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READY?

You cross a speed bump and feel nothing... while your friend slammed his car to the floor: “now that’s handling!” Which of you is right? 😏

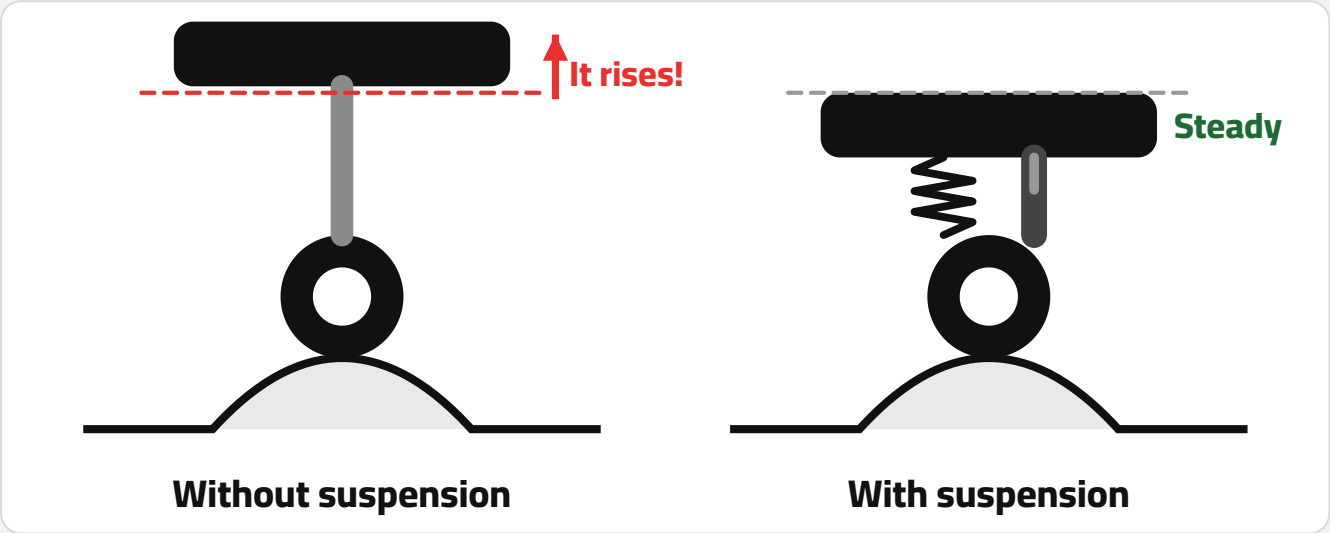
By the end of this chapter you'll know:

- ✓ Why a car needs suspension: grip, comfort, control
- ✓ Every part by name: spring, damper, sway bar, control arms, bushings.
- ✓ Damper vs. strut vs. coilover: what’s the difference?
- ✓ Suspension types: MacPherson, double wishbone, multi-link and the solid axle.
- ✓ Roll, dive and squat: what happens to the car as it moves.
- ✓ Why “too low” loses grip and comfort, and what suits our roads?
- ✓ Signs of worn suspension, and garage talk by its proper names.



01 Why have suspension at all? 🤔

If roads were as flat as mirrors, we wouldn't need suspension. But roads have bumps, potholes and ripples. And a tire only gives you grip **when it actually touches the ground**. That is where the suspension comes in: every part that sits between the tires and the car body.



Simplified illustration: the body stays almost level even when the tire climbs the bump

Suspension has three jobs 🎯

1. Grip

Keeps the tire pressed to the ground, because a tire that's in the air gives you no acceleration, no braking and no steering.

2. Comfort

Absorbs bumps and vibration instead of passing them on to you and the car's parts.

3. Body control

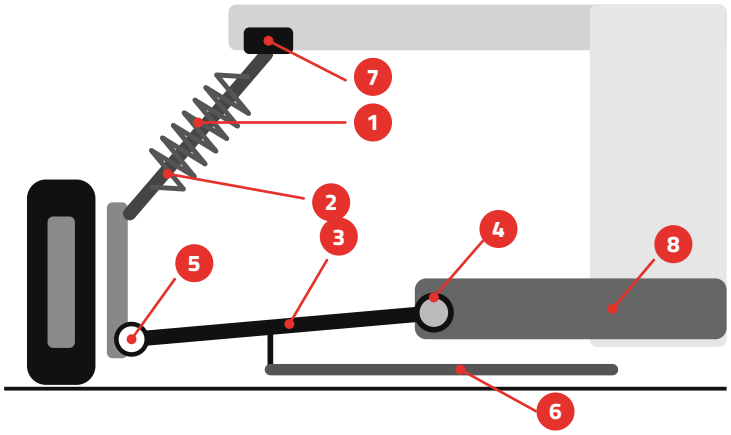
Controls body roll and dive during braking and cornering, and spreads the weight correctly over the tires.

28 M/S metres every second your car covers at 100 km/h. In other words the suspension deals with every bump and pothole in fractions of a second, and never stops for a moment. 😮

⚖️ **Did you know?** A lighter wheel, tire and brakes make it easier for the tire to follow the road. It's called **unsprung weight**, and we'll come back to it in **Chapters 10 and 11**.

02 The suspension map 🗺️

Suspension isn't one part. It is a team of parts, each with its own job. This is one corner (one wheel) of a front suspension. Look at the numbers and match them to the list:



Simplified illustration; the shape and position of parts vary from car to car

1. Spring
Carries the car's weight and absorbs bumps.

2. Damper / strut
Settles the spring's movement and stops bouncing.

3. Control arm
Connects the wheel to the body and lets it move up and down only.

4. Bushing
A ring of rubber (or another material) that softens the connection and stops noise.

5. Ball joint
A ball-and-socket joint that lets the wheel turn with the steering and move with the suspension.

6. Sway bar
Links the two sides and reduces body roll in corners.

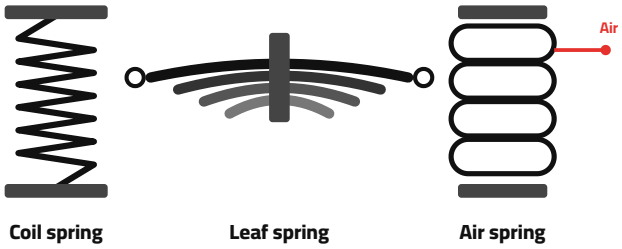
7. Damper top mount
Fixes the damper to the body and absorbs some vibration.

8. Subframe
A metal frame that carries the arms (and sometimes more) and bolts to the body.

375 KG kg roughly at each corner of a car weighing 1,500 kg, if the weight is spread evenly. And all that weight goes through a single spring at each wheel! 💪

03 Springs: the part that carries the weight 🌀

The spring is what **actually carries the car's weight**, and it compresses when you hit a bump and then extends again. Three common types:



Simplified illustrations

Coil: the most common. Simple and light, and it fits around the damper or beside it.

Leaf: curved steel strips stacked on top of each other. Strong, and you see it a lot on vans and pickups.

Air: a rubber bellows filled with compressed air. Its stiffness and height change with air pressure.

Spring rate: how much force does it take to compress it? 📐

Spring rate is the force needed to compress it a set distance, and it's measured in N/mm or kgf/mm in garage talk. A bigger number = a stiffer spring.

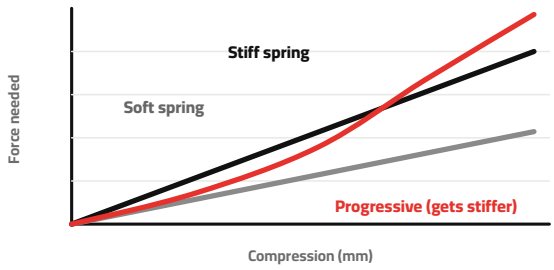


Illustration: each line represents a different spring

1,000 N newtons to compress a spring rated 20 N/mm over a distance of 50 mm (20 × 50), which is about 100 kg of force. And if the distance were only 10 mm, the force would be 200 newtons. 🔧

⚠️ **Careful:** The spring number on its own isn't the whole story. It has to suit the car's weight, the damper and how you use the car.

04 Dampers: the part that calms the spring 🛢️

A spring with no damper = a car that bounces several times after every bump, like a ball. The **damper** stops this: inside it, oil passes through small orifices, so it turns the spring's motion into **heat** and calms it.

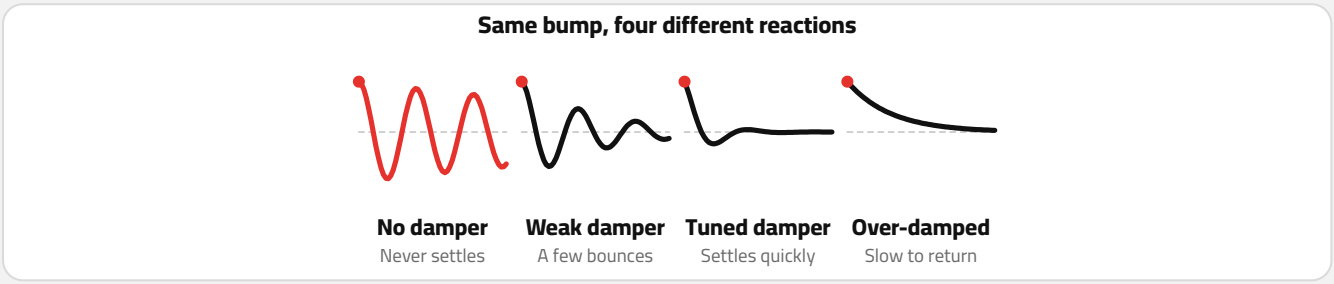


Illustration: the line shows the wheel's movement after the bump

0 KG kg of the car's weight are carried by the damper! The spring does the carrying, and the damper is just a brake on motion, so don't imagine it "strengthens" the car. 🌊

Compression and rebound: 2 in 1 ↕️

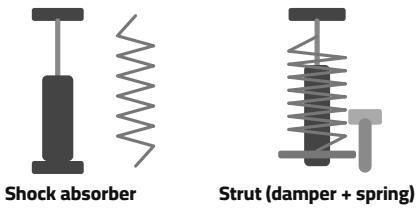
Compression

The wheel goes up (a bump). The damper slows the movement so it doesn't slam into the body.

Rebound

The spring extends and pushes the wheel back. The damper slows the return so it doesn't bounce.

Shock absorber vs. strut



Simplified illustration

The shock absorber: only calms the motion, and the spring sits beside it.

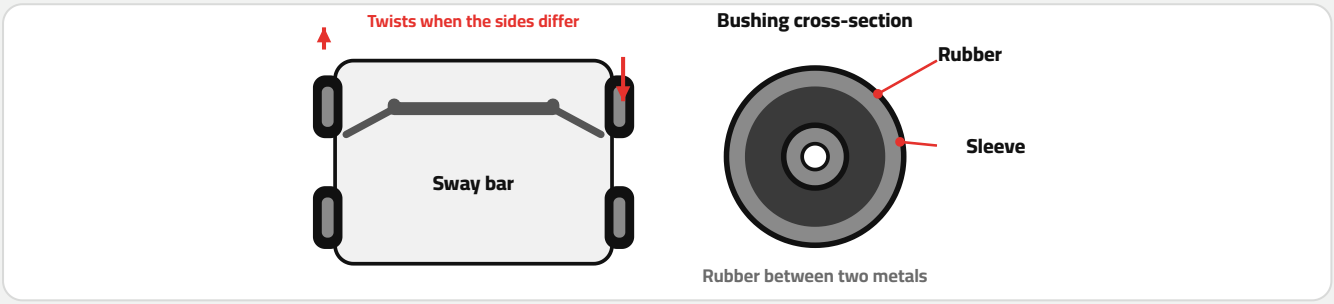
The strut: combines the damper with the spring and is part of the suspension structure, so you can't do without it.

Many dampers contain nitrogen gas that reduces foaming.

05 The rest of the team: arms, bushings and joints 🔩

The sway bar (Sway / Anti-Roll Bar)

A U-shaped steel rod linking the right wheel to the left. When you enter a corner, the outside is compressed and the inside extends, so the bar twists, resists that difference and reduces body roll. On a straight bump (both sides moving together) it barely does anything.



Simplified illustration: the sway bar (left) and a bushing cross-section (right)

🔄 **Link to driving:** If the front bar is relatively stiffer than the rear, the car leans more toward Understeer. If the rear one is stiffer, it leans more toward Oversteer. That is a general rule and it varies from car to car. In **Chapter 6** you saw how the drive layout plays with the same balance.

Control arm

An arm (or more) that connects the wheel to the body and sets its path of movement. Called a “maqass” (scissors) or a “wishbone” because it is sometimes shaped like a V or an A.

Bushing

A ring of rubber (or polyurethane in modified cars) between parts. Rubber softens and stops noise, while stiffer material gives more precision, noise and harshness.

Ball joint

A ball-and-socket joint between the arm and the wheel. It wears with time, giving clunks and slop, and it affects steering.

Subframe

A metal frame the arms bolt to (and sometimes the engine and steering). Usually isolated from the body with rubber.

Damper top mount

Joins the strut to the body, and on the front it has a bearing that lets the damper turn with the steering.

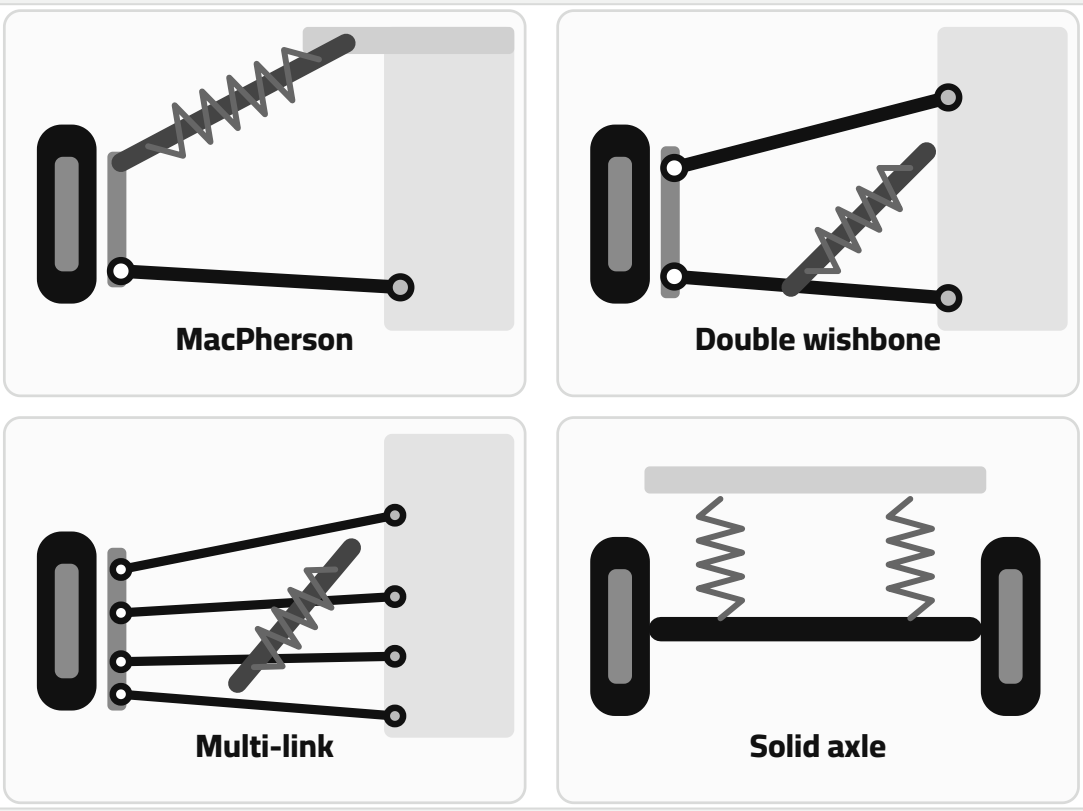
Bump stop

A rubber stopper that keeps the spring and damper from colliding when the suspension compresses all the way.

✅ **DE.AUTO tip:** Bushings tire out before they break. If you hear a clunk or the steering feels “lost”, check them before fitting any new part on top of worn ones.

06 Suspension layouts in brief 🏗️

Manufacturers build suspension in more than one way, and the choice balances cost, space and performance. These are the four best-known layouts (very simplified front views, with the tire on the left):



Positions and lengths are approximate; the aim is just to get the idea

MacPherson (Strut)
One strut + a lower arm. Simple, cheap and space-saving, which is why it is very common on the front.

Double wishbone
Two arms (upper and lower) control the wheel's motion more precisely. You find it on sports and luxury cars.

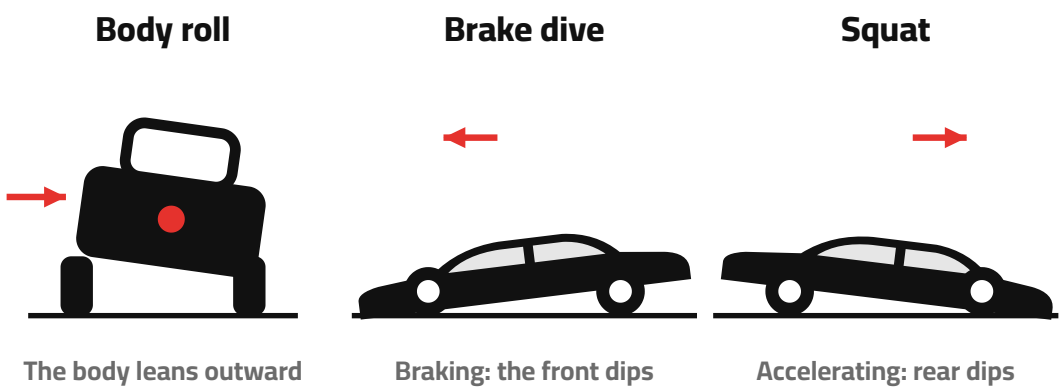
Multi-link
3 to 5 arms, each with its own job. High flexibility for comfort and stability, but more complex and more expensive.

Solid axle (Solid / Live Axle)
One beam connects the two wheels. Strong, simple and good with loads, but the two wheels affect each other.

🧠 **Honestly:** There is no design that is "best" every time. A solid axle, for example, is excellent for load-carrying vehicles and rough roads, while multi-link often gives nicer comfort and stability in passenger cars. Smart design beats a big name.

07 Roll, dive and squat 🌊

A car's weight doesn't stay put. **Weight transfers**: forward when you brake, backward when you accelerate, and outward when you turn. The suspension decides how much it transfers and how fast.



An exaggerated drawing so you can see the idea. The red arrow shows the direction of weight transfer, and the front of the car is on the left

Body roll
The body leans outward in a corner. Stiffer springs and a stronger sway bar reduce it.

Brake dive
The front drops under braking, so more weight shifts onto the front tires.

Squat
The rear drops under hard acceleration, so the front gets lighter.

Weight transfer
The general name for all of this. It decides which tire carries more at any moment.

3 main directions the car's weight moves toward: **front, rear and sideways**. Every tire's load goes up or down depending on your actions. That is what changes the grip moment by moment.

So how does it relate to sliding? 🔄

The tire that carries more weight grips more (up to a point), and the one that gets lighter loses grip. So when weight transfers, the front/rear balance changes: the front doesn't turn the way you want (Understeer) or the rear slides (Oversteer). In **Chapter 6** you saw the role of the drive layout, and here we add the role of springs, dampers and bars.

> 🎯 **The truth:** A little roll isn't a flaw, it gives you a feel for the car's limits. The goal is control, not stiffness.

08 Height and stiffness: the big trade-off ⚖️

Ride height is the suspension mod people love most, and the one whose costs they forget the most. Here are three common routes:

Option	The idea	The good	The catch
Lowering springs	Shorter (and usually stiffer) springs on the stock dampers	Cheaper and simpler	The stock damper may not suit them
Coilover	Damper + spring in one unit, with adjustable height	More precise adjustment if the quality is good	Quality varies, and it may be too harsh for the street
Air suspension	Bags and a compressor, with height at the press of a button	Lifts for bumps and drops for parking	More complex and more expensive, with maintenance and leaks

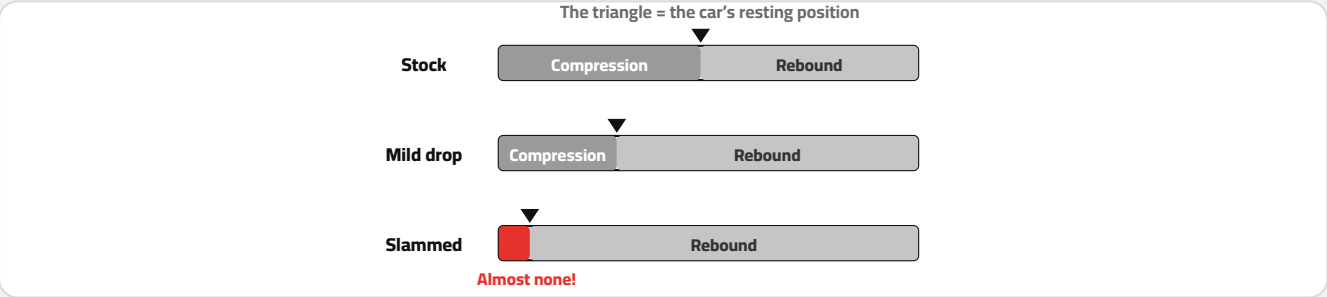


Illustration: the lower you drop the car, the less compression travel remains

Why does "too low" hurt handling? 📉

You run out of suspension travel: the tire reaches the bump stop quickly, so you feel a thump and lose grip.

The geometry changes: the arms leave their designed angles, so wheel angles and steering change as the suspension moves, which hurts stability especially on uneven roads.

Damage to the car: the oil sump, exhaust and bumpers get close to the ground and hit on bumps, and the tires wear unevenly.

✕ **MYTH**
The lower you drop the car, the more stable it gets.

✓ **FACT**
A small, well-thought-out drop may help, but beyond a certain point it becomes a loss in stability and comfort. And cutting the original springs to lower the car is **dangerous**, so don't do it.

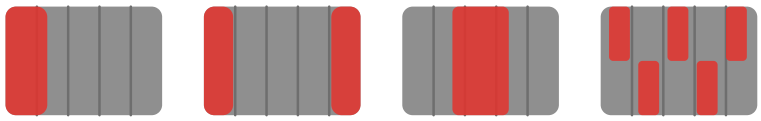
⚖️ **Legally:** Ride height is tied to local traffic regulations and may affect the periodic inspection and insurance. Check the official rules before any modification.

09

Signs of wear, and our roads 🔎

How do you know the suspension needs checking?

Sign	Likely cause	Action
Clunking over bumps	Worn bushings, sway bar links or mounts	Have them checked
Car bounces after a bump	Weak dampers (press down on the wing and let go: if it rebounds more than twice, get it checked)	Replace both sides together
Oil on the damper body	A leaking seal	The damper is usually finished
Squeaks, groans and slop	Ball joints or bushings	Check before it gets worse
Tires wearing oddly	Alignment, dampers or air pressure	Alignment and a check
Car pulls to one side	Alignment, pressure, tire or brakes	Check: it isn't always the suspension



One edge Angles off | **Both edges** Low pressure | **Centre** Too much air | **Cupping** Shocks/balance

Wear spots in red, all just first clues that need confirming by a specialist

Our roads have their own demands 🌵

Speed bumps

Slow down before the bump, go over it straight and gently. Repeated hits wear the joints, bushings and dampers.

Heat

Speeds up the aging of rubber and oil. Check the suspension regularly before summer.

Dust and sand

They eat the rubber dust boots and get into the joints. A torn boot = replace it early.

Potholes and curbs

A hard curb strike can knock the wheel angles out and damage a joint or an arm. If you hear a noise afterwards, get it checked.

10 Extras, technology and a teaser 🧪

Strut tower braces

A metal bar connecting the two front damper towers above the engine. The idea is to strengthen the link between the two sides and reduce the body's flex a little in corners.

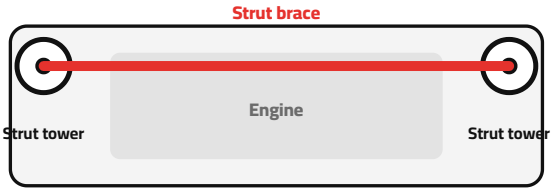


Illustration: top view of the front of the car

> 📏 **No exaggeration:** A brace's benefit depends on how rigid the car's body is, and its effect is usually small. A small improvement, not a miracle.

Smart, adjustable suspension 🤖

Some modern cars have **adaptive dampers**: sensors and a computer change the damper's stiffness within fractions of a second to suit the road and your driving style. Some also have air height adjustment, or driving modes such as Comfort or Sport that change the response at the press of a button.

✕ MYTH	✓ FACT
Stiffer dampers and springs are always better for driving.	Stiffer reduces roll but loses grip on uneven roads, so the tire skips instead of sticking. For the street the most suitable is usually a balance between comfort and control.

The three angles: coming up 📐

After the suspension works, the wheel angles must be set right, or everything is lost. There are only three angles you need to know:

Camber
The tilt of the tire inward or outward when seen from the front.

Toe
The direction of the two tires seen from above: pointing toward each other or apart.

Caster
The tilt of the steering axis forward or backward.

➡️ SOON We go through all of them in **Chapter 10: Tires and Wheels**!

11 Garage talk vs. the proper name 🗣️

If you walk into a garage and they tell you “the shocks and rotils are tired”, this dictionary translates it. Names differ from region to region:

Gulf garage talk	Proper name	What it means
Musaadat	Shock absorber / damper	Slows the spring’s movement
Kuwilat / zunburk / sousa	Coil springs	They carry the car’s weight (not the ignition coils from Chapter 1)
Strut	Strut	Damper + spring + part of the structure
Coilover	Coilover	Adjustable spring and damper in one unit
Maqassat	Control arms / wishbones	They connect the wheel to the body
Bushat / jilab	Bushings	Rubber that isolates and gives flexibility
Rotil	Ball joint	A ball-and-socket joint
Ras dirkson	Tie rod end	Part of the steering system, not the suspension
Dirkson	Steering system	The rack is part of it
Sway bar / stabiliser	Anti-roll bar (sway bar)	Reduces body roll
Manfakh / air bag	Air spring	Carries the weight with air pressure
Rebound rubber	Bump stop	Prevents a collision when the suspension compresses fully
Mismar / bar link	Sway bar link	Connects the bar to the suspension
Shock “chairs”	Top mounts	Connect the damper to the body
Dirkson adjustment / angles	Wheel alignment	Sets the camber, toe and caster

🎯 Test yourself!

Common mistakes 🚧

❌ Replacing just one damper. Always replace them in pairs (right and left on the same axle), or the car behaves differently between the two sides.

❌ Fitting new mod parts on worn bushings and joints. A good part can't fix the worn parts around it.

❌ Forgetting the alignment after any change in height or suspension parts, which wears out the tires and ruins stability.

And here are five questions. Answer without going back to the pages, and the answers are in the wrap-up.

🎯 Challenge 1 **Which part actually carries the car's weight?**

A) The damper | B) The spring | C) The sway bar

🎯 Challenge 2 **The damper's main job?**

A) Calms the spring's movement and stops bouncing | B) Lifts the car | C) Reduces the car's weight

🎯 Challenge 3 **When does the sway bar work most?**

A) On a straight bump | B) When parked | C) In corners, when the two sides move differently

🎯 Challenge 4 **What is the main problem with lowering the car as far as it will go?**

A) Suspension travel shrinks and its geometry changes | B) It gets lighter | C) Comfort goes up

🎯 Challenge 5 **What is the difference between a damper and a strut?**

A) No difference | B) The strut is smaller | C) The strut combines the damper and spring and is part of the suspension structure

Wrap-up 📌

Now you know suspension isn't just "soft bumps". It is a system that keeps the tires pressed to the ground, controls the body's movement, and balances comfort against grip. Springs carry, dampers calm, sway bars reduce roll, and arms and bushings set the path. And every modification gives on one side and takes on the other. 🚀

🎯 Challenge answers:

1) B: The spring.

2) A: Calms the spring's movement and stops bouncing.

3) C: In corners, when the two sides move differently.

4) A: Suspension travel shrinks and its geometry changes.

5) C: The strut combines the damper and spring and is part of the suspension structure.

➡️ But let's be honest: all that grip is worth nothing if you can't stop. In **Chapter 9: Brakes and Safety** we open up the calipers and pads, see how they handle heat, and what the safety systems do to protect you.